

System Dynamics: Part I

ORSA Study text



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Sources:

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System Dynamics

INTRODUCTION

System dynamics is a methodology used to understand and model complex systems and their interactions over time. It focuses on the feedback loops, time delays, and nonlinear relationships that characterize dynamic systems. Developed by Jay Forrester in the 1950s, this approach helps in analyzing how different components of a system influence each other and how they evolve over time.

Key concepts in system dynamics include:

1. ***Stocks and Flows***: Stocks represent the accumulations of resources (like population, inventory, etc.), while flows are the rates of change that affect these stocks.
2. ***Feedback Loops***: These are circular causal pathways where an output of a system loops back to affect its input. Positive feedback amplifies changes, while negative feedback stabilizes the system.
3. ***Time Delays***: Delays in the response of a system can lead to unexpected behaviors and oscillations, making it essential to consider them in modeling.
4. ***Nonlinear Relationships***: Many systems exhibit nonlinear behavior, meaning that small changes in one part of the system can lead to significant changes elsewhere.

System dynamics is widely used in various fields, including economics, environmental science, healthcare, and organizational management, to simulate scenarios, predict outcomes, and inform decision-making. Through visual modeling and simulation, it allows stakeholders to explore the consequences of different policies and strategies over time.

VENSIM

Vensim is a software tool used for system dynamics modeling, which helps in understanding and analyzing complex systems over time. It was developed by Ventana Systems and is widely used in various fields, including business, environmental science, and social sciences.

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Vensim allows users to create graphical models that represent the relationships between different variables and how they influence one another. It provides features for simulation, analysis, and optimization, making it useful for decision-making processes. Users can build models to visualize feedback loops, time delays, and nonlinear interactions, which are essential for studying dynamic systems. The software supports both casual users and advanced modelers, offering a range of functionalities from simple diagramming to complex simulations. Vensim is particularly valuable in scenarios where understanding the behaviour of systems over time is crucial, such as in policy analysis, resource management, and strategic planning.

VENSIM PLE

Vensim PLE is a version of Vensim that has been designed to lower the barriers to the beginning system dynamics modeler. This version is used at ORSA tutorials, it is important to download it also for home preparation of students (self-tests, exam test, lectures, tutorials).

Vensim PLE is fully functional system dynamics software that is free for personal and educational use, and comes complete with sample models, help engine, and Adobe Acrobat format PLE User's Guide. You can download Vensim PLE here: <https://vensim.com/free-downloads/>.

Vensim PLE:

- Is free for academic and personal use.
- Has simplified menus and dialogs.
- Contains fewer option settings.
- Has a fixed tool set.
- Contains fewer model-building tools.
- Contains fewer of functions.

The result is an easy-to-use package that makes it fun to start exploring system dynamics.

Please note that if you require simulation or analysis features not present in Vensim PLE, then you should try the other releases: Vensim PLE Plus, Vensim Professional, or Vensim DSS.

If you are using Vensim PLE for commercial purposes, you may evaluate it for up to 90 days free of charge. If you want to keep using it, you may purchase an inexpensive commercial license in the Vensim Store.

Vensim PLE Documentation (User's Guide) is available at: <https://vensim.com/documentation/index.html>.

To install Vensim PLE, you will need to read and accept a License Agreement.

STOCK AND FLOW DIAGRAMS

The Stock and Flow Diagram (SFD), also known as the Forrester Diagram, is a characteristic diagram in System Dynamics. It is a translation of the causal diagram to a terminology that makes it easy to write equations on a computer. Basically it is a reclassification of elements.

There are no strict rules about how to make this transformation but there is a generally accepted way of tackling the process. The steps to follow are:

1. Take a mental snapshot of the system and the elements included in it (people, sq kms, litres, animals, etc.).

These are its **stocks**.

2. Find or create the elements that constitute the 'stock variation' (persons/day, litres/hour, etc.). These are the **flows**.

3. The remaining elements are the **auxiliary variables**.

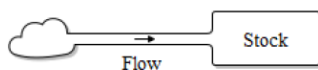
This will do as a general guide. Later on you can make alterations and those stocks that are going to remain constant (e.g., sq m of a room) can be defined as an auxiliary variable instead of stock, which makes the diagram simpler.

That is the whole procedure. Now let us take a closer look at how these elements are represented.

The 'stocks' are those elements that show the state of the model at each instant - they present an accumulation and only vary as a function of other elements called 'flows'. The 'clouds' in a flow diagram are stocks with inexhaustible content. Stocks are represented by a rectangle. e.g.: people, sq km, litres,...



Flows are elements that can be defined as time functions. They can be regarded as representing decisions taken in the system which determine the variations of the stocks. e.g.: people/day, sq km/year,



'Auxiliary variables' and 'constants' allow a better visualisation of the aspects that affect the behaviour of flows, e.g.: density, weight,...

The magnitudes of the flows (\$/day, people/hour, km/year,...) are transmitted through 'material channels'.

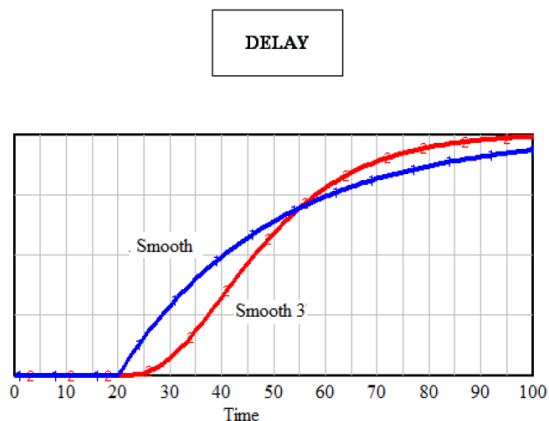


There are also "information links" that transmit information that is not stored.



Finally, we should define the 'delays', which simulate the time taken to convey material or information. In socio-economic systems, there are often delays in the transmission of information and material and they have an important effect on the behaviour of the system.

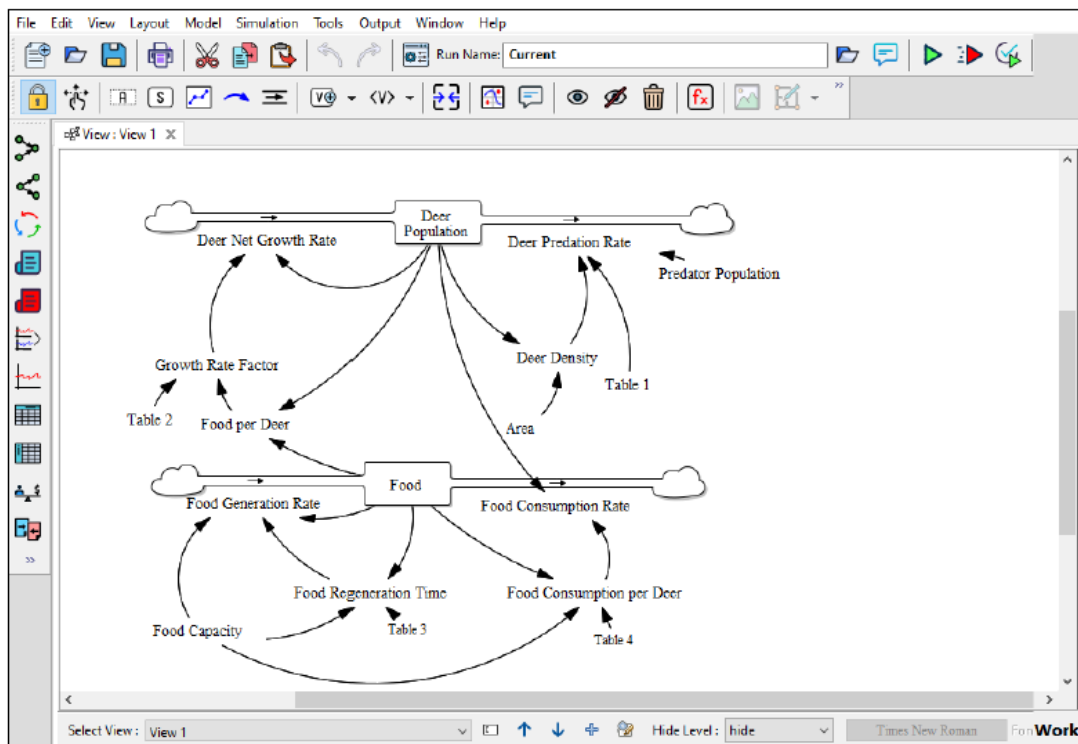
For delays in material we have the functions DELAY3 and SMOOTH. For information delays we use DELAY1 and SMOOTH3. First-order delays (DELAY1 and SMOOTH) with respect to a step input will produce an exponentially asymptotic curve, whereas a third-order delay (DELAY3 and SMOOTH3) leads to a sigmoidal curve. To some extent, information delays act as smoothing filters for the input variable.



COMPUTER SIMULATION IN VENSIM

In this step, we write the concise instructions (i.e., equations) for the computer to interpret our vision of the system. e.g.:

Food Generation Rate = (Food Capacity-Food)/Food Regeneration Time



You have to give numerical values to system variables, functions and tables. This is one of the many ways in which System Dynamics differs from most traditional modelling methods. For example, in econometrics, most of the research effort is dedicated to determining the exact value of the parameters that characterise the system under study.

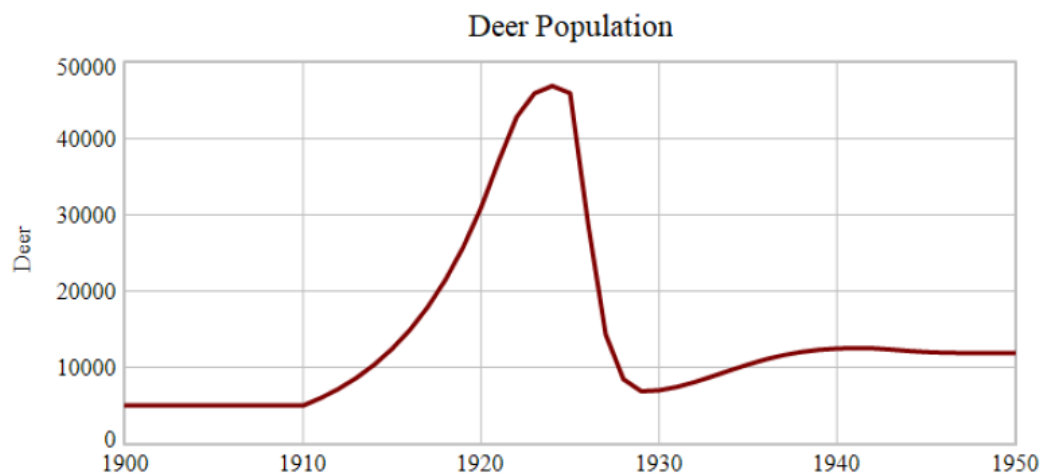
However, in System Dynamics the parameters are only calculated to the necessary degree of exactitude for the model to fulfil its purpose. Given that social systems tend to be fairly sensitive to changes in the values of the parameters, it isn't necessary to dedicate a great deal of time to calculating them exactly.

We can work from a set of approximate values with a view to obtaining an initial idea of the behaviour of the model. Later on we will use sensitivity analyses to identify the relatively small number of parameters whose values have a significant effect on the behaviour of the model or its responses to different policies. In this way we'll discover which parameters are to be calculated exactly. The advantages this method provides as regards savings in time and effort are obvious.

BEHAVIOUR OF THE MODEL

Once the equations have been fed into the computer we can obtain an output in the form of the variation over time of the parameters that we have given it.

We can also compare the behaviour of the model with reality, as the output provided by the model enables us to verify the truth of our hypotheses. On the basis of the difference between the model and reality, we have to reconsider our initial hypotheses and make the necessary adjustments to the model.



A dynamic system possesses a number of aspects that can be evaluated, including:

- Its ability to reproduce the historical data of the system being modelled, under normal and extreme conditions;
- The acceptability of the assumptions made when defining the model; and
- The plausibility of the numerical values adopted for the parameters of the model.

Of course, the first criterion will only be of importance when the others are also verified, as there will be an infinite number of models that are capable of suitably reproducing the historical data of the system without being related to the mechanisms that make up the structure of that system.

Our judgement as to how a model fulfils the above criteria should not be restricted to considering the quantitative information available, since most of the relevant knowledge about social systems is qualitative and held by experts in the field we are considering.

It is important to bear in mind that a model that passes all the evaluation tests is neither an incontestable description of reality nor the only model.

Working from a given set of initial conditions, we determine the evolution of each of the model's variables over the time horizon chosen for the evaluation and record this evolution by means of graphs. By comparing these graphs with their corresponding historical data, we can find out whether the main characteristics of the real behaviour are fulfilled.

This evaluation will then serve as the basis for perfecting the model, by correcting the defects that have been detected and introducing any improvements that are considered appropriate. This reformulated model is then used to carry out a new simulation followed by a further analysis and evaluation. This process is repeated until the model is considered to satisfy the set objectives to a sufficient degree, or until the result of the changes we could introduce would not be worth the effort.

As we said earlier, the first simulation is done with a set of parameter values and initial conditions that are not necessarily exact. Sometimes, in the absence of data, these values will be based on the opinions of experts in the field of the system being studied. As a result, they won't be very exact but often they will nonetheless be useful.

ANALYSIS OF THE RESULTS

Once the model yields an output that is coherent with the past and the present situation, we can simulate the impact of policies or decisions that will lead us to the solution of the problem. It is also possible to locate key factors, by means of sensitivity analyses.

The final model should be relatively simple, so we will have to make the necessary additions and any possible simplifications in order to make the resulting model understandable for the receiver and manageable for the user. On top of the great deal of time and effort required to obtain a complex model, it can be just as complicated as the reality it represents, which hinders communication and means that the model is of very little help in solving the problem at hand.

It should be stressed that although a System Dynamics model has the appearance of a complex set of mathematical equations that might look like it could make a perfect prediction of the future, this is not what we are after. What we are trying to do is increase our knowledge of the system we are studying and create a useful tool for analysing policies which should be judged on the basis of the overall trends they generate.

This difference between the appearance of the model and its intention tends to deceive users who often think they're dealing with a different sort of tool. This possibility recedes when the model is regarded as what it really is - the explicit representation of a mental model. This perception is easier to grasp if the creator of the model keeps the use of inaccessible symbols to a minimum, clarifies the mathematical equations with graphics that are easily understood by non-specialists, and accompanies them with a consistent justification, in other words, if transparency is one of the premises for the creation of the model.

Moreover, the limitations of the model must be made clear, particularly when it is used to predict the impact that different policies would have on the system. We must not forget that the evolution we obtain of the behaviour of the model is based on certain hypotheses regarding the present structure of the system and the circumstances that make up its environment. Therefore, the conclusions drawn will be valid only insofar as the fundamental structure of the system or the outside circumstances that might influence its sensitive parts remain unchanged, and only if we can assume that the model is a correct portrayal of the present.

CONTROL QUESTIONNAIRE

a. Give some examples of **STOCKS**. These are elements where there is an accumulation, such as: persons, dogs, books, etc.

b. Give some examples of **FLOWS**. These are those elements that make Stocks vary. Their units always have a time component, such as litres/year, persons/hour, etc. Note: Flows are not ratios, such as persons/hectare or price/kilo.

c. Give some examples of **AUXILIARY VARIABLES**, which are all the elements that are not Stocks or Flows, such as the price of a kilo of bread or animals per acre.